

连接传统信号处理与边缘人工智能： 感知与计算约束下的助听设备语音增强

Bridging Conventional Signal Processing and Edge AI: Speech Enhancement for Hearing Devices Under Perceptual and Computational Constraints

李明爽 / Mingshuang Li

mingshuang.li@csun.edu

*Department of Communication Disorders and Sciences, California State University,
Northridge, 18111 Nordhoff Street, Northridge, CA 91330, USA*

Abstract

Prescriptive amplification restores audibility, yet many hearing-aid users continue to struggle in noise because amplification does not resolve target-to-masker competition. Conventional noise reduction improves comfort and effort more reliably than intelligibility, whereas spectral and formant enhancement approaches strengthen selected speech cues. Deep learning can produce larger laboratory gains, but translation to wearable devices requires causal, real-time processing within tight memory, computation, and battery budgets. This review connects evidence from conventional signal processing, cue-directed enhancement, human-listener studies, and hardware implementations. It argues that hearing-device algorithms should be judged by perceptual benefit within a stated device budget. Targeted formant enhancement illustrates interpretable auditory-informed processing, while compact neural models may support target identification, scene analysis, and control. A likely near-term direction is a hybrid architecture in which compact AI guides efficient, controlled signal processing.

Keywords: Speech enhancement; conventional signal processing; artificial intelligence; hearing aids.

1. Introduction

Modern hearing aids can restore audibility with considerable precision, yet speech in noise remains difficult for many users. Amplification cannot fully compensate for reduced frequency selectivity, temporal and binaural deficits, or the cognitive demands of following one talker among competing voices. Reverberation, moving sound sources, changing communication goals, and source-related degradation such as speech muffling add to the difficulty in everyday listening.

Prescriptive amplification, typically implemented through frequency- and level-dependent gain and WDRC, provides the foundation of hearing-aid processing by restoring audibility within the listener's reduced dynamic range (Keidser et al., 2011). Its purpose, however, is not to separate target speech from competing sound. When speech and noise overlap, amplification may make both more audible without inherently improving the target-to-masker ratio, and restored audibility alone may be insufficient at moderate-to-high noise levels. Speech enhancement addresses this complementary problem by improving access to the target under adverse acoustic conditions. It is best understood as an adaptive processing layer within the broader hearing-aid signal chain, operating alongside rather than replacing prescriptive amplification (Kayser et al., 2022).

Speech enhancement has approached this problem in several ways. Digital noise reduction (DNR) attenuates components judged to be noise, whereas spectral enhancement strengthens selected features of the target. More recently, deep neural networks have learned mappings from noisy mixtures to masks, filters, spectra, or waveforms. The perceptual consequences differ across these approaches. Noise reduction may improve comfort without improving recognition, and cue enhancement may increase the salience of selected speech information without changing the overall signal-to-noise ratio (SNR). Either approach can reduce naturalness or environmental awareness when applied too aggressively.

Deep learning has expanded what is technically possible, but offline performance says little by itself about suitability for a hearing aid. A wearable device must process sound continuously and with little delay, usually without relying on a remote server, while sharing limited computation, memory, and battery capacity with the rest of the hearing-aid system. It must also preserve loudness comfort, spatial cues, and access to relevant background sounds. In this review, Edge AI refers to causal or streaming inference performed on the device or closely coupled local hardware rather than in the cloud. The practical issue is whether AI can deliver reliable listener benefit throughout the day, in unfamiliar acoustic scenes, within the limits of the device that the listener wears.

Previous reviews have examined DNR, deep-learning speech enhancement, and AI in hearing aids (Bentler and Chiou, 2006; Andersen et al., 2021; Healy et al., 2023). The present review takes a narrower translational view. It considers what conventional signal processing has taught us about listener outcomes, what cue-targeted enhancement adds, how far neural systems have progressed, and where device constraints still limit translation. The final section considers how conventional signal processing and Edge AI may be integrated, using targeted formant enhancement as one interpretable test case rather than a complete solution.

2. Scope and Evaluation Framework

This integrative narrative review examines representative work on conventional DNR, spectral and formant enhancement, and AI-based speech enhancement for hearing devices. It is selective rather than exhaustive, with emphasis on foundational studies that established important perceptual principles and recent work providing human-listener evidence, ecological evaluation, causal or streaming operation, or measurements on resource-constrained hardware. A comprehensive taxonomy of neural architecture, microphone-array processing, dereverberation, and target-speaker extraction is beyond its scope.

The evidence is considered along two dimensions: perceptual value and device feasibility. Perceptual outcomes include speech recognition, listening effort, sound quality, preference, and environmental awareness, which should not be treated as interchangeable. Device feasibility includes causal and real-time operation, execution on or near the device, latency, memory, computation, energy use, and integration with the broader hearing-aid signal chain. Objective measures such as STOI, HASPI, PESQ, and SI-SDR remain useful during development, but they do not substitute for listener testing. For hearing devices, an improvement is meaningful only when it produces reliable listener benefit within a clearly stated device budget.

3. Conventional Signal Processing Approaches to Speech Enhancement

Conventional speech-enhancement algorithms have generally followed two broad strategies: attenuating components attributed to competing noise and strengthening perceptually relevant features of the target speech. These strategies address different failure modes, and their perceptual record provides a useful baseline for evaluating AI-based methods.

3.1 Digital noise reduction

Conventional single-channel DNR estimates whether time-frequency regions are dominated by speech or noise and reduces gain where noise is judged to dominate. Implementations have used modulation analysis, spectral subtraction, Wiener filtering, statistical noise tracking, and related gain-control methods (Bentler and Chiou, 2006). Because speech and noise overlap, attenuation can remove low-level speech components as well as the masker, while rapid gain changes can introduce audible artifacts.

The perceptual record is mixed. Conventional DNR has improved comfort and acceptable noise level more reliably than intelligibility. Laboratory and field studies found limited recognition benefit, and a meta-analysis of adult hearing-aid studies reached a similar conclusion (Bentler et al., 2008; Lakshmi et al., 2021). Even without an intelligibility gain, noise reduction may reduce listening effort or release cognitive resources (Sarampalis et al., 2009). The recurring pattern of improved comfort or reduced effort without better recognition remains an important benchmark for AI-based systems.

3.2 Spectral and formant enhancement: From broad contrast modification to targeted cues

Spectral enhancement was motivated by the loss of frequency selectivity associated with sensorineural hearing loss. Early methods narrowed formant bandwidths, amplified spectral peaks, attenuated valleys, or expanded contrast across multiple channels. Their effects were usually small or condition dependent. Some studies reported modest improvements in consonant or sentence recognition, whereas others found no intelligibility benefit or poorer sound quality (Bunnell, 1990; Simpson et al., 1990; Stone and Moore, 1992; Baer et al., 1993; Franck et al., 1999). Dynamic enhancement and individualized parameters produced larger gains in selected noise conditions, but the effect did not generalize broadly (Chen et al., 2012, 2013). Spectral enhancement has also failed when tested with spectrally sharpened sentences, underscoring that greater contrast does not automatically create more usable speech information (Nitttrouer et al., 2015).

Formant-directed processing has a longer history than is sometimes recognized. Boers (1980) described formant enhancement for listeners with sensorineural hearing loss, and Summerfield et al. (1985) examined formant-bandwidth narrowing. Narrowing did not improve consonant place identification in quiet, although broadening was harmful. A later approach, contrast-enhancing frequency shaping (CEFS), increased high-frequency emphasis as a function of hearing loss and the speech spectrum. CEFS improved the auditory-nerve representation of F2 in animals with noise-induced hearing loss (Miller et al., 1999). Bruce (2004) extended the scheme to running speech and examined its compatibility with multiband compression using an auditory-periphery model. This work supplied a physiological rationale for formant-directed enhancement, but not evidence of improved perception in hearing-aid users.

A related series of behavioral studies focused specifically on F2 and, more recently, combined F2-F3 enhancement. F2 enhancement improved vowel-formant discrimination for listeners with hearing loss and produced condition-dependent benefits for older listeners and for vowel and consonant identification in noise (Woodall and Liu, 2013; Guan and Liu, 2019; Li, 2021). Extending this work to sentences, Li and Liu (2026) compared unmodified, F2-enhanced, and F2-F3-enhanced natural and N95-muffled speech. Enhancement provided no measurable benefit in quiet, whereas both approaches improved recognition overall in four-talker babble. F2-F3 enhancement was more consistent across speaking

conditions and SNRs, particularly when speech muffling and external masking occurred together. This pattern suggests a benefit from restoring broader mid- to high-frequency spectral structure, although the specific contribution of F3 could not be isolated.

The studies also point to an important boundary condition: benefit is most likely when the relevant cues are degraded, because favorable conditions leave little room for improvement. Enhancement can alter spectral tilt, formant-amplitude relations, or sound quality, and most published formant studies processed the target before it was mixed with noise even though a hearing device receives the mixture. The evidence therefore demonstrates perceptual efficacy under controlled signal access rather than readiness for real-time deployment. Formant enhancement remains useful as an interpretable test bed for selective cue modification, but its computational demand, latency, memory, and energy use have yet to be measured on target hardware.

4. AI-Based Approaches to Speech Enhancement

In practical devices, AI-based enhancement enters an established hearing-aid signal chain in which prescriptive amplification, WDRC, directionality, feedback control, and output limiting remain essential. Learned models can add estimation, separation, and adaptive control under conditions that are difficult for fixed-rule processing. The relevant question is therefore not whether a neural model performs well offline, but what useful role it can sustain within the complete device.

4.1 Neural methods and human-listener evidence

Machine-learning enhancement differs from classical estimation because the mapping from noisy input to desired output is learned from examples. Early supervised systems estimated ideal binary or ratio masks; later networks estimated continuous masks, complex spectra, adaptive filters, or time-domain waveforms (Wang and Chen, 2018). The first human-listener demonstrations produced unusually large intelligibility gains, but training and test conditions were closely matched and future frames were often available (Healy et al., 2013).

Subsequent work addressed increasingly realistic constraints. Training variation expanded across talkers, noises, corpora, recording channels, reverberation, and languages. Causal speaker-separation and dereverberation systems retained perceptual benefit while avoiding future information, although conversion to real-time-capable operation carried a cost (Healy et al., 2021). Cross-corpus studies also showed that generalization to one kind of novelty does not guarantee performance in another (Pandey and Wang, 2020). By 2023, a causal attentive recurrent network trained with large numbers of talkers and noises produced substantial average intelligibility improvements for listeners with hearing loss under varied test conditions (Healy et al., 2023). Other work has demonstrated benefit for hearing-aid users rather than relying only on normal-hearing simulations (Diehl et al., 2023).

Human-listener studies now show that neural enhancement can improve communication, although benefit remains tied to sound quality, failure modes, and the relation between training data and everyday use. Causality alone does not ensure deployability: a model may still require a long analysis window, excessive memory, or more computation than a wearable device can sustain.

4.2 Commercial devices and everyday outcomes

Recent studies have begun to evaluate commercially implemented systems. Christensen et al. (2024) compared premium hearing aids during everyday wear and found that average satisfaction was similar across devices, although

satisfaction with the DNN-based system was less dependent on ambient SNR. Fitzgerald et al. (2025) combined electroacoustic measurements, clinical speech-in-noise tests, and ecological ratings; benefits were evident, but their magnitude depended on the test and masker. Overby et al. (2025) showed that listeners' preference for DNN noise reduction also depends on how it is combined with wide dynamic-range compression (WDRC).

Folkeard et al. (2026) reached a related conclusion using commercial hearing aids configured with different combinations of directionality and noise reduction. The best overall results occurred when DNN processing was combined with beamforming, and benefits varied with masker type and spatial configuration. These interactions make it difficult to evaluate an AI module in isolation: beamforming changes the signal presented to the neural processor, and WDRC may preserve or magnify residual artifacts.

For a hearing aid, the desired output is not necessarily the closest approximation to an anechoic clean signal. It is the signal that best supports the listener's current goal while remaining natural, comfortable, spatially coherent, and informative about the surroundings. This is a different objective from maximum noise suppression.

5. Why AI for Hearing Aids Is a Distinct Engineering Problem

5.1 From offline models to real-time operation under resource constraints

Many neural enhancement systems are evaluated offline on desktop or server-class hardware, whereas a hearing aid must process a continuous audio stream on the device and without dependable access to remote computation. Offline evaluation, causal processing, real-time execution, and untethered operation are not equivalent. A causal model uses no future input, yet analysis windows, buffering, memory, or computation may still produce unacceptable delay or exceed the available processor. End-to-end delay also includes analysis and synthesis, look-ahead, and data transfer; open fittings can be particularly unforgiving (Goehring et al., 2018). Faster-than-real-time execution on an external processor is therefore insufficient evidence of hearing-aid feasibility.

Computation and energy are coupled because enhancement runs for hours from a small battery while sharing resources with the rest of the signal chain. Parameter count alone is a poor proxy for deployability: working memory, numerical precision, data movement, duty cycle, and front- and back-end processing also affect runtime and energy use. TinyLSTMs illustrated how pruning, quantization, and state-update skipping can reduce computation on a microcontroller (Fedorov et al., 2020). Recent hardware studies reported a 28-nm processor consuming 740 microwatts with 39.96 ms total latency and streaming enhancement on a programmable behind-the-ear hearable (Park et al., 2025; Itani et al., 2025). These results establish feasibility, but power and latency must still be interpreted in relation to form factor, battery capacity, expected wear time, duty cycle, and measurement boundaries.

5.2 System integration and device-level evaluation

Enhancement must operate alongside directionality, feedback control, frequency shaping, WDRC, output limiting, and binaural processing. Device-level comparisons should add candidate enhancement strategies to the same appropriately fitted amplification baseline, so that observed differences reflect enhancement rather than audibility or compression. Recognition, effort, and quality should then be considered together with latency, memory, computation, and energy. The relevant endpoint is listener benefit under the constraints of the intended device; Table 1 summarizes the information needed to assess it.

Table 1. Key information for evaluating the perceptual benefit and device feasibility of speech-enhancement systems.

Domain	Information to report
Signal processing and latency	Sample rate; microphone channels; frame and hop; look-ahead; causal status; analysis-synthesis and end-to-end delay
Model and computation	Architecture; parameter count; numerical precision; operations per frame or second; front-end and reconstruction costs
Memory, hardware, and energy	Model storage; peak working memory; target processor; runtime; measured power or energy; included and excluded stages
Perceptual evaluation	Population and audiograms; speech, masker, SNR, and reverberation; recognition; effort; quality; artifacts; spatial awareness
System integration	Beamforming; WDRC; fitting formula; feedback control; binaural processing; cloud, smartphone, or on-device execution
Generalization and ecological validity	Unseen noise, talker, corpus, room, language, device, and recording channel; field or ecological-momentary evidence

6. Future Directions: Integrating Conventional Signal Processing and Edge AI

Future progress is likely to depend less on choosing between conventional signal processing and learned processing than on assigning each component the role it performs best. Prescriptive amplification will remain the foundation of hearing-loss compensation, while speech enhancement may combine efficient conventional signal processing with compact learned models for scene analysis, target estimation, and adaptive control. Within this hybrid framework, four developments appear especially important.

Near-term systems are likely to remain modular. Prescriptive frequency shaping and WDRC will continue to provide hearing-loss compensation, while beamforming, feedback control, and other conventional signal-processing components solve well-defined problems efficiently. Learned systems can complement these components without displacing them with a single end-to-end model. Hardware-software-audiology co-design will matter as much as model architecture: an algorithm that improves an offline score but consumes resources needed by other functions or shortens usable battery life is not necessarily the better hearing-aid solution.

Learned processing may be most useful upstream of acoustic modification. A compact causal model could identify the current target, classify the scene, estimate relevant cue trajectories, and judge its own confidence. A bounded processor could then apply gain, filtering, or cue enhancement. The learned output would remain low dimensional and the acoustic change interpretable. When confidence is low, the device could reduce processing strength or return to a conservative mode rather than fail abruptly.

Evaluation will also need to move beyond a single leaderboard score. Objective measures remain useful for screening, but hearing-device studies need human outcomes and device measurements from the same implementation. A smaller intelligibility gain may be preferable if it preserves quality and spatial awareness, avoids objectionable delay, and can run throughout the day. Comparisons should include the complete processing chain rather than attributing the result to one module.

Personalization is likely to become more contextual and more closely tied to user intent. Audiogram-conditioned processing is only a starting point. Users differ in listening goals, tolerance for suppression, language experience, and the

value they place on environmental sound. Continuous on-device learning may be costly and unstable; a more practical near-term approach is a compact pretrained model with a small number of listener-specific controls and occasional preference updates. The wanted talker may change as the user turns the head, and in traffic the safest output may not be the one with the highest SNR.

Targeted formant enhancement provides an interpretable test case because both the manipulated cues and their perceptual consequences can be specified. The next step is mixture-based F2/F3 estimation with bounded enhancement and individualized WDRC, evaluated in terms of tracking accuracy, listener outcomes, computation, memory, latency, and energy. Its value should be assessed on the same hardware against conventional and full-band neural systems, progressing from unfamiliar acoustic conditions to listeners with clinically representative hearing loss and field use. Such comparisons would clarify the balance between perceptual benefit and practical feasibility and could also be extended to other auditory-informed cue modifications.

7. Conclusions

Prescriptive amplification will remain the foundation of hearing-aid processing, but it does not by itself resolve target-to-masker competition. Speech enhancement complements rather than replaces prescriptive amplification, and future systems are likely to integrate conventional signal processing with Edge AI. Evidence from conventional DNR distinguishes improvements in comfort from gains in intelligibility, while spectral and formant enhancement approaches show that selective cue modification can help under specific forms of degradation. Deep learning offers broader estimation and separation capabilities, although its practical value still depends on generalization, sound quality, latency, memory, computation, energy, and integration with the complete hearing-aid system. A hybrid architecture is a credible near-term direction: compact AI can identify the target, characterize the scene, and guide processing, while bounded auditory-informed methods implement the acoustic modification. Progress should therefore be judged by whether an algorithm provides reliable listener benefit while meeting clearly specified constraints on computation, memory, latency, and energy use.

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Author Declarations

Conflict of Interest

The author declares no conflicts of interest related to this manuscript.

Ethics Approval

Not applicable.

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Correspondence Author

Mingshuang Li, Ph.D.

E-mail: mingshuang.li@csun.edu